Simultaneous exposure of non-diabetics to high levels of dioxins and mercury increases their risk of insulin resistance.

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Insulin resistance and the defective function of pancreatic β-cells can occur several years before the development of type 2 diabetes. It is necessary to investigate and clarify the integrated effects of moderate-to-high exposure to dioxins and mercury on the pancreatic endocrine function. This cross-sectional study investigated 1449 non-diabetic residents near a deserted pentachlorophenol and chloralkali factory. Metabolic syndrome related factors were measured to examine associations with serum dioxin and blood mercury. We also investigated associations between insulin resistance (HOMA-IR > 75th percentile), defective pancreatic β-cells function (HOMA β-cell > 75th percentile), serum dioxins and blood mercury. After adjusting for confounding factors, we found that insulin resistance increased with serum dioxins ($b = 0.13$, $P < 0.001$) and blood mercury ($b = 0.01$, $P < 0.001$). Moreover, participants with higher serum dioxins or blood mercury were at a significantly increasing risk for insulin resistance ($P(trend) < 0.001$). The joint highest tertile of serum dioxins and blood mercury was associated with elevated HOMA-IR at 11 times the odds of the joint lowest tertile (AOR 11.00, 95% CI: 4.87, 26.63). We hypothesize that simultaneous exposure to dioxins and mercury heightens the risk of insulin resistance more than does individual exposure.